

HUJI Applicative Research Report — Week 40, 2026

3 high-potential paper(s) · Weekly · Week 40, 2026 · Generated September 28, 2026

Hebrew University researchers have unveiled breakthroughs with significant commercial implications across biotechnology and agriculture. One innovation introduces a novel cell-penetrating peptide nucleic acid for efficient intracellular drug delivery, potentially revolutionizing gene therapies by simplifying administration. Another provides an advanced mouse model for more accurate preclinical testing of anti-aging therapies. Finally, a discovery in wheat genetics promises new avenues for developing climate-resilient, high-yield crops essential for global food security.

RESEARCHERS & RESEARCH SUBJECTS — HIGH-POTENTIAL PAPERS

- **Eylon Yavin** — Drug Discovery, Genomics, Synthetic Biology
- **Yehuda Tzfati** — Drug Discovery, Genomics, Synthetic Biology, Clinical
- **Zvi Peleg** — AgriTech, Genomics, Synthetic Biology

ALL BRANCHES — RESEARCH HIGHLIGHTS

1. Multiple Charged Purine (MCP) PNA as a Simple Mode for Cellular Uptake. 33/50

"Simple cellular uptake for genetic drugs: a versatile platform without complex delivery vehicles."

Main researcher: Eylon Yavin

The Institute for Drug Research, The School of Pharmacy, The Faculty of Medicine, The Hebrew University of Jerusalem, Hadassah Ein-Kerem, Jerusalem 9112102, Israel.

Published in ACS polymers Au · 2026-06

ABOUT THIS RESEARCH

Researchers developed a new type of synthetic DNA analogue called MCP PNA that can naturally enter cells without needing complex delivery vehicles. This solves a major historical bottleneck where these stable molecules were too large or poorly charged to cross cell membranes.

COMMERCIAL ANGLE

Development of a versatile, high-stability platform for intracellular genetic targeting and antisense therapeutics without the need for expensive lipid nanoparticles or viral vectors.

COMMERCIALISATION METRICS

Novelty

6/10

The work provides a practical chemical modification to improve PNA uptake, but utilizing charge modification for cellular delivery is a known strategy in oligonucleotide chemistry rather than a groundbreaking platform shift.

Commercial Potential

8/10

This represents a platform-enabling technology that solves a fundamental bottleneck (cellular uptake) for PNA, significantly expanding the addressable market for PNA-based therapeutics and diagnostics.

Market Size

9/10

The work addresses a fundamental delivery bottleneck for PNA, a platform with massive potential across antisense therapy, diagnostics, and gene editing markets. Solving cellular uptake enables a wide range of therapeutic applications, making this a platform-enabling advancement with a very large total addressable market.

Tech Readiness

3/10

The work focuses on fundamental chemical modification of PNA for cellular uptake, representing basic proof-of-concept laboratory research without evidence of in vivo efficacy or a developed delivery platform.

IP Strength

7/10

The introduction of a specific chemical modification (MCP) to solve a fundamental PNA delivery bottleneck is highly patentable as a composition of matter. While it enables a platform for cellular uptake, its ultimate strength depends on the breadth of the claims regarding the purine modifications.

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2. Addendum: Telomouse-a mouse model with human-length telomeres generated by a single amino acid change in RTEL1.

32/50

"A high-fidelity human aging model to accelerate anti-aging and longevity drug development."

Main researcher: Yehuda Tzfati · tzfati@mail.huji.ac.il.

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Published in Nature communications · 2026-05

ABOUT THIS RESEARCH

Researchers have developed a specialized mouse model that possesses human-like telomere lengths through a precise single amino acid modification in the RTEL1 protein. This model allows for more accurate preclinical testing of aging-related diseases and therapies by better mimicking human cellular biology.

COMMERCIAL ANGLE

The creation of a high-fidelity biological surrogate for human aging enables pharmaceutical companies to accelerate the development and validation of anti-aging and longevity drugs.

COMMERCIALISATION METRICS

Novelty

8/10

The research demonstrates a highly innovative genetic engineering approach by using a single amino acid substitution to recapitulate a complex human phenotype, creating a powerful, platform-enabling model for aging and telomere biology.

Commercial Potential

6/10

While the creation of a humanized telomere mouse model is a valuable tool for aging and oncology research, its primary value is as a specialized research reagent rather than a broad-market therapeutic or platform technology.

Market Size

7/10

A mouse model with human-length telomeres serves as a high-value platform for aging and oncology research, potentially tapping into the massive biologics and longevity drug discovery markets. However, as a specialized research tool, its market is primarily limited to academic and pharmaceutical R&D budgets rather than direct clinical application.

Tech Readiness

3/10

The work describes the creation of a fundamental biological research tool (a transgenic mouse model) rather than a therapeutic or diagnostic product. While it enables downstream drug discovery, the technology itself is at the basic research/proof-of-concept stage for modeling human telomere biology.

IP Strength

8/10

The creation of a transgenic animal model via a specific single amino acid mutation in a highly conserved gene (RTEL1) represents a platform-enabling tool with strong potential for patentability regarding the specific genetic construct and its utility in humanized telomere research.

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3. Antagonistic pleiotropy at the stem solidness 1 locus balances wheat stem architecture and yield components under terminal drought.

31/50

"Climate-resilient wheat: a genetic discovery to maintain yield under drought conditions."

Main researcher: Zvi Peleg · zvi.peleg@mail.huji.ac.il.

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Published in TAG. Theoretical and applied genetics. Theoretische und angewandte Genetik · 2026-08

ABOUT THIS RESEARCH

Researchers identified a specific genetic locus (stem solidness 1) in wheat that controls the trade-off between stem structural strength and grain yield. The study explains how adjusting this genetic trait can help wheat plants survive terminal drought without sacrificing overall productivity.

COMMERCIAL ANGLE

Development of climate-resilient, high-yield wheat cultivars through precision breeding or CRISPR gene editing targeting the s1 locus.

COMMERCIALISATION METRICS

Novelty

5/10

The identification of a specific locus controlling stem architecture and its trade-off with yield under drought is a valuable incremental contribution, but follows established paradigms of QTL mapping and antagonistic pleiotropy in crop science.

Commercial Potential

7/10

The identification of a specific locus controlling the trade-off between stem strength and yield under drought provides a direct target for marker-assisted breeding or CRISPR editing in wheat, a global staple crop. While highly relevant for seed commercialization, the score is tempered by the inherent complexity of 'antagonistic pleiotropy' which may require precise tuning rather than a simple binary switch.

Market Size

9/10

Wheat is one of the world's most critical staple crops; a genetic solution to maintain yield under terminal drought addresses a massive global food security market with high industrial demand.

Tech Readiness

3/10

The research focuses on identifying a genetic locus and the biological trade-off (antagonistic pleiotropy) between stem architecture and yield, which is fundamental discovery research (TRL 2-3) rather than a developed seed product or breeding protocol ready for commercial deployment.

IP Strength

7/10

The identification of a specific genetic locus (stem solidness 1) governing a trade-off between architecture and yield provides a clear target for Marker-Assisted Selection (MAS) or CRISPR editing, which is highly patentable in agricultural biotech.

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